



AI-SUPPORTED RESCHEDULING FOR TRAIN DISPATCHERS

TEAM: LADIES ON TRACK — FIVE PEOPLE. ONE PRODUCT.

JOHANNA Idea Driver - Research - Original concept - Rail operations research - Pitch & presentation	PAULINA Research - Interface - Rail operations research - Digital Twin - Interface Development	KARINA Research - Interface - Rail operations research - Interface Design & UX - Presentation & Storytelling	VIDUSHI Builder - AI Agents Logic - Recommendations Logic - Interface Development	DIKSHA Builder - AI Agents Logic - Recommendations Logic - Deliverables & Storytelling
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PROJECT IDEA & CHALLENGE CONTEXT

41.3%

of DB long-distance trains were delayed in H1 2026.
(Deutsche Bahn, Interim Report 2026)

High-Pressure Decision Making & Cascading Impact: Dispatchers must make complex decisions under time pressure, assessing consequences spreading far beyond local disruptions. A locally reasonable decision can create network-wide consequences: missed connections, stranded passengers, and people who stop taking the train.

What Train the Train Does: A real-time decision-support app leveraging multiple agents to evaluate rescheduling options and minimise overall network impact. Features real-time network overview, conflict detection with alerts, AI-supported conflict resolution, and full human control.

IMPLEMENTATION DETAILS & SYSTEM ARCHITECTURE

1. Live Data Integration & Pilot Station - Ingests timetable and real-time disruption data via the Deutsche Bahn API. - Start Small: Optimizes platform arrivals, departures, and track occupancy at one station (one Control Center) for fast deployment and immediate local impact.	2. Conflict Detection & Multi-Agent Evaluation - Train Network Conflict Detection & Alerts: Real-time network overview of arriving and departing trains with automated disruption alerts. - Agent 1 (Network): Minimises total delay across the network. - Agent 2 (Passengers): Minimises the number of passengers affected at any moment.
3. Orchestration Agent & Human in the Loop - Orchestration Agent: Evaluates and provides ranked recommendations based on options provided by both specialized agents. - Human in the Loop: The dispatcher reviews AI recommendations and predicted impact, making the final decision with full human authority.	4. Decentralized & Locally Hosted by Design - Connect & Scale: Decentralized agent network where regional independent agents communicate with neighbor agents to improve the overall rail system. - Infrastructure Resilience: High resilience against attacks/outages with no single point of failure; enables optimized local solutions.

TOOL STACK

Tool / Technology	Category / Area	Role & Implementation Purpose
Bilt	Application Framework & UI	Used to build the core application interface and structure, establish visual direction and color scheme, and support the dispatcher user experience.
Codex	Code Optimization	Used for refining, debugging, and adapting generated logic and multi-agent backend scripts.
Google Gemini	Visuals & Asset Generation	Used for ideation, generating visuals, branding graphics, and project asset generation throughout development.

VISION & FUTURE SCOPE

Connect & Scale: Decentralized Network Scale from single-station control to regional independent agents that communicate horizontally with neighbor agents. Delivers resilience against attacks, optimizes local solutions, and accommodates provider diversity.	Predict & Improve: Disturbance Prediction Leverage historical data to forecast bottlenecks and delays before disruptions occur. Enables dispatchers to take earlier preventative actions, significantly reducing operational network delays.
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